

A stadium spectral effect algebra with non-singleton atomic faces

Counterexample packet for arXiv:1811.12407

9 August 2026

Status. Likely-valid full negative answers to open questions 1 and 5, pending human review. The counterexample is finite-dimensional and explicit.

1 Questions answered

Jenčová and Plávala ask in Questions 1 and 5 on p. 11 of [1] whether the extreme states of a spectral effect algebra have the atomic exposed form, and whether

$$\hat{a} = \{s \in \mathcal{S}(E) : s(a) = 1\}$$

is a singleton for every sharp one-dimensional effect $a \in S_1(E)$. Both answers are *no*.

2 The stadium and its affine effect algebra

Let D be the Euclidean unit disk in $\mathbb{R}^2$, let $L = [-1, 1] \times \{0\}$, and set

$$K = L + D.$$

Thus K is the planar stadium

$$K = \left\{ (x_1, x_2) : (\max\{|x_1| - 1, 0\})^2 + x_2^2 \leq 1 \right\}. \quad (1)$$

It is a compact centrally symmetric convex body.

Let $E(K)$ be the convex effect algebra of all affine functions $f : K \rightarrow [0, 1]$, with unit the constant function 1. As recalled in [1], its state space is affinely isomorphic to K : a point $x \in K$ acts by evaluation, $s_x(f) = f(x)$.

For $z = (z_1, z_2) \in (\mathbb{R}^2)^*$, the support norm of K is

$$\|z\|_* = \max_{x \in K} z(x) = |z_1| + \sqrt{z_1^2 + z_2^2}. \quad (2)$$

Indeed support functions add under Minkowski sums. This norm is strictly convex: equality in its triangle inequality forces equality in the Euclidean triangle inequality, hence positive collinearity; two unit vectors with that property are equal.

7 Conclusions and open questions

In this article we have proved that there can be either only one or uncountably many contexts contained in a spectral effect algebra as well as few other results concerning spectral effect algebras. We believe that spectral effect algebras provide a suitable framework for the study of operational theories possessing some kind of spectral properties and therefore are an interesting research direction for quantum foundations. In any case, there are still quite a few open questions that will be left for future work.

1. What are the extreme points of a state space of a spectral effect algebra? In Prop. 5 we have shown that all exposed points of the state space of a spectral effect algebra have the form $s = \hat{a}$ for $a \in S_1(E)$. It is an open question whether one can show a similar result for the extreme points of the state space.
2. Let $A = \{a_1, \dots, a_n\}$ be a context and let $\hat{A} = \text{conv}(\cup_i \hat{a}_i)$. Since $\hat{a}_i$ are affinely independent, $\hat{A} = \oplus_c \hat{a}_i$ is their convex direct sum. Let I be a set that indexes all of the contexts A_α of an effect algebra E ; does it hold that $\mathfrak{S}(E) = \cup_{\alpha \in I} \hat{A}_\alpha$? If $\hat{a}$ are E -exposed points for all $a \in S_1(E)$, this would be even a stronger result than the one proposed in the question above.
3. When is the cardinality of contexts always the same? It is straightforward to see that all contexts in the effect algebras used in quantum theory have the same number of elements. It would be very interesting to know when this is true in general.
4. Is it possible to extend the results of Prop. 4 to a result similar to Prop. 10? The property in question is to show that $\sum_{i=1}^n \mu_i a_i = f = \sum_{j=1}^m \nu_j b_j$ where $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_m\}$ are contexts would imply that for every μ_i there is ν_j such that $\mu_i = \nu_j$. A stronger version of said result would be to show that Prop. 10 holds for all spectral effect algebras; this can be either done by showing that the state space of every spectral algebra is sharply determining or by other means.
5. For $a \in \mathfrak{S}_1(E)$, does $\hat{a}$ contain only one state? It is rather easy to find convex effect algebras where the sets $\hat{a}$ are not singletons, one can for example consider the effect algebra corresponding to the square state space (but note that this effect algebra is not spectral).

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Figure 1: The source paper's open questions 1–5 [1, p. 11].

3 Why the effect algebra is spectral

We record the elementary mechanism in a form that also verifies every effect-algebra detail needed by the counterexample.

Lemma 1 (Two-context spectrality). *Let K be a compact centrally symmetric convex body whose support norm $\|\cdot\|_*$ on the dual space is strictly convex. Then $E(K)$ is a spectral effect algebra. More precisely, for every $\|y\|_* = 1$,*

$$p_y(x) = \frac{1 + y(x)}{2}, \quad p_{-y} = 1 - p_y$$

are sharp one-dimensional effects, so $\{p_y, p_{-y}\}$ is a context, and every effect is a linear combination of such a context with coefficients in $[0, 1]$.

Proof. Identify an affine function with a pair (α, z) via $f(x) = \alpha + z(x)$. Central symmetry gives

$$f \geq 0 \iff \alpha \geq \|z\|_*, \quad 0 \leq f \leq 1 \iff \|z\|_* \leq \min\{\alpha, 1 - \alpha\}.$$

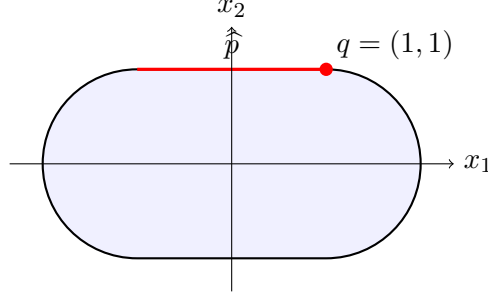


Figure 2: The stadium K . The red top segment is the atomic face $\widehat{p}$; its endpoint q is extreme but not exposed.

Fix $\|y\|_* = 1$ and suppose $0 \leq g = (\beta, w) \leq p_y = (1/2, y/2)$. Then

$$\|w\|_* \leq \beta, \quad \|y/2 - w\|_* \leq \frac{1}{2} - \beta.$$

Consequently,

$$\frac{1}{2} = \|y/2\|_* \leq \|w\|_* + \|y/2 - w\|_* \leq \beta + \frac{1}{2} - \beta = \frac{1}{2}.$$

All inequalities are equalities. Strict convexity makes w and $y/2 - w$ nonnegative collinear, and the two displayed bounds then give $g = \lambda p_y$ for some $\lambda \in [0, 1]$. Thus p_y is one-dimensional. If also $g \leq 1 - p_y$, choose $x \in K$ with $y(x) = 1$. Then $g(x) = \lambda$ while $(1 - p_y)(x) = 0$, so $\lambda = 0$. Therefore p_y is sharp. The same applies to p_{-y} , and their sum is 1.

Finally let $f = (\alpha, z) \in E(K)$ and put $r = \|z\|_*$. If $r > 0$, set $y = z/r$; if $r = 0$, choose any dual unit vector y . Since $r \leq \min\{\alpha, 1 - \alpha\}$,

$$\mu = \alpha + r, \quad \nu = \alpha - r$$

belong to $[0, 1]$, and direct calculation gives

$$f = \mu p_y + \nu p_{-y}.$$

This is a spectral decomposition over the context $\{p_y, p_{-y}\}$. □

Equation (2) and Lemma 1 show that the stadium effect algebra $E(K)$ is spectral in exactly the finite-context sense used in the source paper.

4 The two negative answers

Theorem 2 (Non-singleton atomic face). *Question 5 of [1] has a negative answer. In the spectral effect algebra $E(K)$, the effect*

$$p(x_1, x_2) = \frac{1 + x_2}{2}$$

belongs to $S_1(E(K))$, but $\widehat{p}$ is not a singleton.

Proof. The functional $y(x_1, x_2) = x_2$ has $\|y\|_* = 1$, so Lemma 1 gives $p = p_y \in S_1(E(K))$. Under the evaluation identification $\mathcal{S}(E(K)) = K$,

$$\widehat{p} = \{x \in K : p(x) = 1\} = [-1, 1] \times \{1\},$$

the whole top line segment of the stadium. □

Theorem 3 (Extreme need not be exposed). *Question 1 of [1] also has a negative answer. The state $q = (1, 1) \in K = \mathcal{S}(E(K))$ is extreme, but it is not exposed and hence cannot have the form $\hat{a} = \{q\}$ for any $a \in S_1(E(K))$.*

Proof. If $q = (x + x')/2$ with $x, x' \in K$, then both second coordinates are at most 1 and average to 1, so both equal 1. Formula (1) then puts both first coordinates in $[-1, 1]$; since their average is 1, both equal 1. Thus $x = x' = q$, proving extremality.

For non-exposedness, write $K = L + D$. If a nonzero functional $z = (z_1, z_2)$ is maximized at $q = (1, 0) + (0, 1)$, then $(0, 1)$ must maximize z over the Euclidean disk. Hence z is a positive multiple of $(0, 1)$. But that functional is maximized on all of $[-1, 1] \times \{1\}$, not uniquely at q . Therefore q is not exposed. If $\hat{a} = \{q\}$ for an effect a , then a itself would expose q by its maximum value 1, a contradiction. $\square$

5 Checks, scope, and novelty

The construction is two-dimensional; no closure, completeness, or infinite-sum issue is hidden. The proof directly checks the source definition of a context and gives a spectral decomposition of every affine effect. The square mentioned in the source fails because its support norm is not strictly convex. Rounding the square's left and right sides into tangent semicircles makes the dual norm strictly convex while deliberately retaining the top and bottom faces.

This packet fully answers Questions 1 and 5 negatively. It does not settle Questions 2–4 or the broad weak-duality Question 6.

A bounded search through 9 August 2026 used the source title and exact question language together with the phrases *spectral effect algebra*, *singleton atomic face*, *centrally symmetric state space*, and *stadium*. No explicit answer by this counterexample was found. Subsequent work of Jenčová and Pulmannová [2] studies centrally symmetric state spaces and characterizes related, but differently formulated, compression-base notions of spectrality. That paper makes the Banach-space geometry strongly adjacent, so novelty confidence is *moderate to low*; the present packet's contribution is the direct finite-context calculation and its explicit application to the two 2019 questions.

Human-review recommendation. Check the equivalence between point evaluations and states of $E(K)$, the one-dimensionality equality case in Lemma 1, and whether [2] or later citing work explicitly records the same answers before making a novelty claim.

References

- [1] A. Jenčová and M. Plávala, *On the properties of spectral effect algebras*, Quantum 3 (2019), 148, doi:10.22331/q-2019-06-03-148; arXiv:1811.12407.
- [2] A. Jenčová and S. Pulmannová, *Geometric and algebraic aspects of spectrality in order unit spaces: a comparison*, J. Math. Anal. Appl. 504 (2021), 125360, doi:10.1016/j.jmaa.2021.125360; arXiv:2102.01628.